Data Communication & Computer Networks — Topic 6: OSI Layers in Detail

Practice Test Booklet — 247 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A — Data Link Layer: Flow & Error Control

Q1. Which sub-layer of the Data Link Layer is responsible for flow control and error control?

Data Link Layer का कौन-सा sub-layer flow control और error control के लिए ज़िम्मेदार होता है?

(A) MAC **(B)** LLC (Logical Link Control)
(C) PVC **(D)** None of the above

[Source: BCI Subject Test-8.pdf] · Easy

Q2. Which of the following functions are performed by the Data Link Layer? (1) Framing (2) Error Detection (3) Flow Control (4) Path Determination

निम्न में से कौन-से कार्य Data Link Layer करता है? (1) Framing (2) Error Detection (3) Flow Control (4) Path Determination

(A) 1, 2 and 3 only **(B)** 1 and 4 only
(C) 2 and 4 only **(D)** 1, 2, 3 and 4

[Source: BCI Subject Test-9.pdf] · Easy

Q3. The Frame Check Sequence (FCS) field in an Ethernet frame is used for:

Ethernet frame में Frame Check Sequence (FCS) field किस काम आता है?

(A) Flow control **(B)** Address resolution
(C) Error detection using CRC **(D)** Medium access

[Source: BCI Subject Test-9.pdf] · Easy

Q4. A Stop-and-Wait ARQ channel has high latency and large bandwidth. What problem occurs?

Stop-and-Wait ARQ channel पर latency ज़्यादा और bandwidth बड़ी हो तो क्या समस्या होती है?

(A) High throughput **(B)** Low channel utilization
(C) CRC failure **(D)** Frame duplication

[Source: BCI Subject Test-8.pdf] · Easy

Q5. A Go-Back-N sender uses window size 4. Frame 2 is lost. What happens after the receiver receives frames 3 and 4?

Go-Back-N sender window size 4 उपयोग करता है. Frame 2 खो जाता है. Receiver को frames 3 और 4 मिलने के बाद क्या होता है?

(A) Receiver stores frames 3 and 4 in buffer **(B)** Receiver sends ACK for frame 4
(C) Receiver discards frames 3 and 4 **(D)** Receiver acts like Selective Repeat

[Source: BCI Subject Test-8.pdf] · Easy

Q6. A noisy channel uses a Hamming code with minimum distance 3. What can it guarantee?

एक noisy channel पर minimum distance 3 वाला Hamming code क्या guarantee दे सकता है?

(A) Detect 2-bit errors **(B)** Correct 2-bit errors
(C) Correct 3-bit errors **(D)** Detect 4-bit errors

[Source: BCI Subject Test-8.pdf] · Easy

Q7. Which of the following verifies the integrity of a message?

निम्न में से कौन-सा message की integrity जाँचता है?

(A) Message Encryption **(B)** Message Checksum
(C) Message Digest **(D)** None of these

[Source: BCI Subject Test-9.pdf] · Easy

Q8. Which of the following statements about MAC and LLC sublayers is TRUE? (1) MAC sublayer handles physical addressing (2) LLC sublayer provides error control and flow control (3) MAC sublayer performs routing

MAC और LLC sub-layers के बारे में कौन-सा कथन सही है? (1) MAC sub-layer physical addressing संभालता है (2) LLC sub-layer error control और flow control देता है (3) MAC sub-layer routing करता है

(A) 1 and 2 only **(B)** 2 and 3 only
(C) 1 and 3 only **(D)** 1, 2 and 3

[Source: BCI Subject Test-9.pdf] · Medium

Q9. Which of the following is NOT a responsibility of the LLC sublayer?

निम्न में से कौन-सा LLC sub-layer की ज़िम्मेदारी नहीं है?

(A) Flow control **(B)** Error control
(C) Service interface to Network layer **(D)** Physical addressing

[Source: BCI Subject Test-9.pdf] · Medium

Q10. Which of the following is NOT a function of the Data Link Layer?

निम्न में से कौन-सा Data Link Layer का function नहीं है?

(A) Framing **(B)** Error detection
(C) Error correction **(D)** Routing

[Source: BCI Subject Test-9.pdf + CN Topic Test-04.pdf] · Medium

Q11. Error control is the process of detecting and correcting errors, while _____ is a mechanism to ensure the efficient delivery of data.

Error control, errors को detect और correct करने की प्रक्रिया है, जबकि _____ data के कुशल delivery का mechanism है.

(A) Error control, Flow control **(B)** Process control, Network control
(C) Session control, Flow control **(D)** OSI Control, Data control

[Source: CN Topic Test-02.pdf] · Medium

Q12. _____ control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.

_____ control उन procedures का समूह है जो sender द्वारा acknowledgment की प्रतीक्षा से पहले भेजे जाने वाले data की मात्रा सीमित करते हैं.

(A) Flow **(B)** Error
(C) Transmission **(D)** Access control

[Source: Unit Test -2 Solutions.pdf (OCR), p17] · Medium

Q13. What is the main purpose of the Sliding Window in TCP?

TCP में Sliding Window का मुख्य उद्देश्य क्या है?

(A) Address resolution **(B)** Flow control
(C) Error detection **(D)** Encryption

[Source: CN Topic Test 5.pdf] · Medium

Q14. Flow control prevents which of the following issues?

Flow control निम्न में से किस समस्या को रोकता है?

(A) Deadlock **(B)** Congestion collapse
(C) Overflow of the receiver's buffer **(D)** Packet routing error

[Source: CN Topic Test-04.pdf] · Medium

Q15. In Stop-and-Wait ARQ, after sending one frame, the sender:

Stop-and-Wait ARQ में एक frame भेजने के बाद sender:

(A) Waits for acknowledgment **(B)** Sends the next frame immediately
(C) Stops transmission permanently **(D)** Sends a NAK

[Source: CN Topic Test-04.pdf] · Medium

Q16. In Stop-and-Wait protocol, the sender sends:

Stop-and-Wait protocol में sender भेजता है:

(A) Multiple frames at once **(B)** One frame at a time
(C) Acknowledgments only **(D)** Data without waiting

[Source: CN Topic Test-04.pdf] · Medium

Q17. What is the window size in Stop-and-Wait ARQ?

Stop-and-Wait ARQ में window size कितनी होती है?

(A) 0 **(B)** 1
(C) N **(D)** 2

[Source: CN Topic Test-04.pdf] · Medium

Q18. What is the main disadvantage of Stop-and-Wait flow control?

Stop-and-Wait flow control की मुख्य कमी क्या है?

(A) Complexity **(B)** Low efficiency due to waiting
(C) High memory usage **(D)** Frame loss

[Source: CN Topic Test-04.pdf] · Medium

Q19. In Stop-and-Wait ARQ, what happens when a frame is lost?

Stop-and-Wait ARQ में frame खो जाए तो क्या होता है?

(A) Receiver sends NAK **(B)** Sender retransmits after timeout
(C) Receiver retransmits **(D)** No retransmission

[Source: CN Topic Test-04.pdf] · Medium

Q20. In Go-Back-N ARQ, if an error is detected in a frame:

Go-Back-N ARQ में किसी frame में error मिले तो:

(A) Only that frame is retransmitted **(B)** All frames are ignored
(C) That frame and all subsequent outstanding frames are retransmitted **(D)** No frame is retransmitted

[Source: CN Topic Test-04.pdf] · Medium

Q21. In a Go-Back-N protocol, if the window size = 63, how many sequence number bits are required?

Go-Back-N protocol में window size = 63 हो तो sequence number के कितने bits चाहिए?

(A) 5 **(B)** 6
(C) 7 **(D)** 8

[Source: BCI Subject Test-8.pdf] · Medium

Q22. A Selective Repeat sliding window protocol uses a window size of 7 with 3-bit sequence numbers. Is it valid?

Selective Repeat sliding window protocol में 3-bit sequence numbers के साथ window size 7 क्या वैध है?

(A) Valid **(B)** Invalid – violates the maximum window rule
(C) Valid only for piggybacking **(D)** Valid only with Selective Repeat

[Source: BCI Subject Test-8.pdf] · Medium

Q23. Which protocol has better bandwidth utilization?

किस protocol का bandwidth utilization बेहतर होता है?

(A) Stop-and-Wait **(B)** Selective Repeat
(C) Go-Back-N **(D)** None of the above

[Source: CN Topic Test-04.pdf] · Medium

Q24. What is the Hamming distance between the two codes 10010010 and 11011001?

दो codes 10010010 और 11011001 के बीच Hamming distance कितनी है?

(A) 3 **(B)** 4
(C) 6 **(D)** 2

[Source: Unit Test -2 Solutions.pdf (OCR), p28] · Medium

Q25. The Hamming distance between two code words can easily be found if we apply the _____ operation.

दो code words के बीच Hamming distance आसानी से मिल जाती है यदि हम _____ operation लगाएँ.

(A) XOR **(B)** XNOR
(C) OR **(D)** AND

[Source: Unit Test -2 Solutions.pdf (OCR), p35] · Medium

Q26. To detect or correct errors, we send extra bits with the data. These extra bits are called:

Errors detect या correct करने के लिए data के साथ भेजे गए अतिरिक्त bits कहलाते हैं:

(A) redundancy **(B)** correction
(C) detection **(D)** parity

[Source: Unit Test -2 Solutions.pdf (OCR), p36] · Medium

Q27. Which error detection method uses one extra bit added to the end of a byte?

कौन-सी error detection method में byte के अंत में एक अतिरिक्त bit जोड़ी जाती है?

(A) CRC **(B)** Checksum
(C) Hamming code **(D)** Parity bit

[Source: CN Topic Test-04.pdf] · Medium

Q28. Which error detection method consists of a parity bit for each data unit as well as an entire data unit of parity bits?

किस error detection method में हर data unit के लिए एक parity bit के साथ-साथ parity bits का पूरा data unit भी होता है?

(A) Simple parity check **(B)** CRC
(C) Two-dimensional parity check **(D)** Checksum

[Source: Unit Test -2 Solutions.pdf (OCR), p7] · Medium

Q29. A simple parity-check code can detect:

एक simple parity-check code detect कर सकता है:

(A) odd number of errors **(B)** even number of errors
(C) positive number of errors **(D)** negative number of errors

[Source: Unit Test -2 Solutions.pdf (OCR), p34] · Medium

Q30. Which of the following is an error detection technique?

निम्न में से कौन-सी error detection technique है?

(A) ARQ **(B)** Parity Bit
(C) Stop-and-Wait **(D)** Sliding Window

[Source: CN Topic Test-04.pdf] · Medium

Q31. High-level Data Link Control (HDLC) is a protocol for communication over point-to-point and multipoint links. HDLC is a _____ protocol.

High-level Data Link Control (HDLC) point-to-point और multipoint links पर communication का protocol है. HDLC एक _____ protocol है.

(A) byte-oriented **(B)** character-oriented
(C) bit-oriented **(D)** none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p15] · Medium

Q32. The most common protocol for point-to-point access is the Point-to-Point Protocol (PPP), which is a _____ protocol.

Point-to-point access का सबसे सामान्य protocol Point-to-Point Protocol (PPP) है, जो एक _____ protocol है.

(A) bit-oriented **(B)** byte-oriented
(C) character-oriented **(D)** none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p16] · Medium

Q33. Which protocol allows the sender to transmit multiple frames before needing an acknowledgment?

कौन-सा protocol sender को acknowledgment की ज़रूरत से पहले कई frames भेजने देता है?

(A) Stop-and-Wait **(B)** Go-Back-N
(C) ALOHA **(D)** CSMA/CD

[Source: CN Topic Test-04.pdf] · Medium

Q34. In CRC, if the received codeword is divided by the generator and the remainder is zero, the receiver concludes:

CRC में यदि remainder शून्य हो, तो receiver निष्कर्ष निकालता है:

(A) Error is present / त्रुटि है **(B)** No error detected / कोई त्रुटि नहीं मिली
(C) Frame is lost / Frame खो गया **(D)** ACK is corrupt / ACK ख़राब है

[Source: Practice — original (textbook-based)] · Medium

Q35. In Selective Repeat ARQ, when a frame is corrupted, the sender retransmits:

Selective Repeat ARQ में frame ख़राब होने पर दोबारा भेजता है:

(A) All outstanding frames / सभी लंबित frames **(B)** Only the corrupted frame / केवल ख़राब frame
(C) No frame / कोई नहीं **(D)** The last frame only / केवल अंतिम frame

[Source: Practice — original (textbook-based)] · Medium

Q36. Parity check can detect which type of errors?

Parity check कौन-सी errors detect कर सकता है?

(A) Any number of errors / कितनी भी errors **(B)** Odd number of bit errors / विषम संख्या में bit errors
(C) Only burst errors / केवल burst errors **(D)** No errors / कोई नहीं

[Source: Practice — original (textbook-based)] · Medium

Q37. Forward Error Correction (FEC) differs from ARQ in that:

Forward Error Correction (FEC), ARQ से भिन्न है क्योंकि:

(A) It needs retransmission / इसे retransmission चाहिए
(B) The receiver corrects errors without retransmission / Receiver बिना retransmission errors ठीक करता है
(C) It cannot detect errors / यह errors detect नहीं करता **(D)** It works only on LAN / यह सिर्फ़ LAN पर चलता है

[Source: Practice — original (textbook-based)] · Medium

Q38. In Go-Back-N ARQ, the receiver's window size is:

Go-Back-N ARQ में receiver की window size होती है:

(A) 1 **(B)** Same as the sender's window / Sender की window के बराबर
(C) 2^(n−1) **(D)** Unbounded / असीमित

[Source: Practice — original (textbook-based)] · Medium

Q39. Delaying an acknowledgement so that it can be attached to the next outgoing data frame in the reverse direction is called:

Acknowledgement को रोककर विपरीत दिशा में जाने वाले अगले data frame के साथ जोड़ना कहलाता है:

(A) Piggybacking **(B)** Broadcasting
(C) Framing **(D)** Stuffing

[Source: Practice — original (textbook-based)] · Medium

Q40. Network congestion occurs when:

Network congestion तब होता है जब:

(A) The offered traffic exceeds the network's capacity / Offered traffic network की क्षमता से अधिक हो **(B)** Two hosts use the same port / दो hosts एक ही port प्रयोग करें
(C) A frame is corrupted by noise / Noise से frame खराब हो **(D)** The DNS server is down / DNS server बंद हो

[Source: Practice — original (textbook-based)] · Medium

Q41. Which of the following techniques uses both acknowledgments and sequence numbers?

निम्न में से कौन-सी technique acknowledgments और sequence numbers दोनों उपयोग करती है?

(A) Go-Back-N **(B)** Stop-and-Wait
(C) Selective Repeat **(D)** All of the above

[Source: Unit Test -2 Solutions.pdf (OCR) + CN Topic Test-04.pdf] · Hard

Q42. In the TCP header, the acknowledgment field value represents:

TCP header में acknowledgment field का मान किसे दर्शाता है?

(A) Number of previous bytes received **(B)** Total number of bytes to receive
(C) Sequence number of the next byte expected **(D)** Sequence of zeros and ones

[Source: CN Topic Test-04.pdf] · Hard

Q43. For Stop-and-Wait ARQ, for 10 data packets sent, ... acknowledgements are needed.

Stop-and-Wait ARQ में 10 data packets भेजने पर ..... acknowledgements चाहिए होते हैं.

(A) 9 **(B)** 20
(C) 11 **(D)** 10

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q44. In the _____ protocol, if no acknowledgment for a frame has arrived, we resend all outstanding frames.

_____ protocol में, किसी frame का acknowledgment न आए तो हम सभी outstanding frames दोबारा भेजते हैं.

(A) Stop-and-Wait ARQ **(B)** Go-Back-N ARQ
(C) Selective-Repeat ARQ **(D)** Both B and C

[Source: Unit Test -2 Solutions.pdf (OCR), p8] · Hard

Q45. In Go-Back-N ARQ, if 6 is the number of bits for the sequence number, then the maximum size of the send window must be:

Go-Back-N ARQ में sequence number के 6 bits हों तो send window का अधिकतम आकार कितना होगा?

(A) 63 **(B)** 61
(C) 31 **(D)** 64

[Source: Unit Test -2 Solutions.pdf (OCR), p9] · Hard

Q46. The maximum number of outstanding frames in Selective Repeat (with N sequence numbers) is:

Selective Repeat में outstanding frames की अधिकतम संख्या (N sequence numbers के साथ) होती है:

(A) N **(B)** N − 1
(C) N/2 **(D)** 2N

[Source: CN Topic Test-04.pdf] · Hard

Q47. Any coding scheme for error control needs to have at least the parameters of:

Error control के लिए किसी भी coding scheme में कम से कम निम्न parameters होने चाहिए:

(A) codeword size **(B)** dataword size
(C) minimum Hamming distance **(D)** All of them

[Source: Unit Test -2 Solutions.pdf (OCR), p37] · Hard

Q48. The message 1001 is to be transmitted using the CRC polynomial $x^3 + x + 1$ to protect it from errors. The codeword that should be transmitted is:

संदेश 1001 को CRC बहुपद $x^3 + x + 1$ से errors से बचाने के लिए भेजना है. भेजा जाने वाला codeword होगा:

(A) 1001100 **(B)** 1001011
(C) 1001110 **(D)** 1001000

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q49. The checksum is used in the Internet by several protocols although not at the _____ layer.

Internet में checksum कई protocols उपयोग करते हैं, हालाँकि _____ layer पर नहीं.

(A) data link **(B)** transport
(C) network **(D)** none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p33] · Hard

Q50. Which of the following is NOT a responsibility of the network layer?

निम्न में से कौन-सा network layer की ज़िम्मेदारी नहीं है?

(A) Congestion control **(B)** Error control
(C) Routing **(D)** Inter-networking

[Source: CN Topic Test-02.pdf] · Hard

Q51. The types of data services offered by an LLC (Logical Link Control) layer are:

LLC (Logical Link Control) layer द्वारा दी जाने वाली data services के प्रकार हैं:

(A) Connectionless (fast and unreliable) **(B)** Connection-oriented (slow and reliable)
(C) Both A and B **(D)** None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p32] · Hard

Q52. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The codeword that should be transmitted is:

संदेश 11001001 को CRC बहुपद $x^3 + 1$ से errors से बचाने के लिए भेजना है. भेजा जाने वाला codeword होगा:

(A) 11001001000 **(B)** 11001001011
(C) 11001010 **(D)** 110010010011

[Source: GATE CS 2007] · Hard

Q53. Let G(x) be the generator polynomial used for CRC checking. What condition must G(x) satisfy to detect an odd number of bits in error?

मान लीजिए G(x), CRC जाँच में उपयोग होने वाला generator polynomial है. Odd संख्या में बिट-त्रुटियाँ पकड़ने के लिए G(x) पर क्या शर्त होनी चाहिए?

(A) G(x) contains more than two terms **(B)** G(x) does not divide 1 + x^k for any k within the frame length
(C) 1 + x is a factor of G(x) **(D)** G(x) has an odd number of terms

[Source: GATE CS 2009] · Hard

Q54. In CRC coding, if the data word is 111111, the divisor is 1010 and the remainder is 110, which of the following is the transmitted code?

CRC coding में, यदि data word 111111 है, divisor 1010 है और remainder 110 है, तो निम्न में से कौन-सा भेजा गया code है?

(A) 011111101 **(B)** 001010110
(C) 111111110 **(D)** 110111111

[Source: UGC NET Computer Science Dec 2022 (via samagracs)] · Hard

Q55. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

एक 2 km लंबे broadcast LAN की bandwidth 10^7 bps है और यह CSMA/CD उपयोग करता है. Signal तार में 2×10^8 m/s से चलता है. इस network पर उपयोग होने वाला न्यूनतम packet आकार कितना है?

(A) 50 bytes **(B)** 100 bytes
(C) 200 bytes **(D)** None of the above

[Source: GATE CS 2003 + RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q56. A network using CSMA/CD has a bandwidth of 20 Mbps. If the maximum propagation time is 25 μs, what is the minimum size of the frame?

CSMA/CD उपयोग करने वाले network की bandwidth 20 Mbps है. यदि अधिकतम propagation time 25 μs हो, तो frame का न्यूनतम आकार क्या होगा?

(A) 500 bits **(B)** 1000 bits
(C) 1500 bits **(D)** 2000 bits

[Source: UGC NET Computer Science Jan 2025 (via samagracs solved paper)] · Hard

Q57. A CSMA/CD network transmits data at 100 Mbps over a 1 km cable with no repeaters. If the minimum frame size required is 1250 bytes, what is the signal speed in the cable (in km/sec)?

एक CSMA/CD network बिना repeaters के 1 km cable पर 100 Mbps से data भेजता है. यदि आवश्यक न्यूनतम frame आकार 1250 bytes हो, तो cable में signal की गति (km/sec में) कितनी है?

(A) 8000 **(B)** 10000
(C) 16000 **(D)** 20000

[Source: GATE CSE 2015 Set 3] · Hard

Q58. In the starting phase (slow start) of the TCP congestion control algorithm, the congestion window size:

TCP congestion control algorithm के शुरुआती चरण (slow start) में congestion window का आकार:

(A) Increases linearly **(B)** Does not increase
(C) Increases exponentially **(D)** Remains constant at threshold

[Source: GATE CS 2008] · Hard

Q59. Consider the following statements: I. TCP connections are full duplex. II. TCP has no option for selective acknowledgment. III. TCP connections are message streams. Which of the following is correct?

निम्न कथनों पर विचार करें: I. TCP connections full duplex होती हैं. II. TCP में selective acknowledgment का कोई विकल्प नहीं है. III. TCP connections message streams हैं. निम्न में से क्या सही है?

(A) Only I is correct **(B)** Only I and II are correct
(C) Only II and III are correct **(D)** All of I, II and III are correct

[Source: GATE CS 2012] · Hard

Q60. The minimum Hamming distance required to detect 2 bit errors is:

2 bit errors detect करने के लिए न्यूनतम Hamming distance चाहिए:

(A) 2 **(B)** 3
(C) 4 **(D)** 5

[Source: Practice — original (textbook-based)] · Hard

Section B — Data Link Layer: Framing & MAC Sub-layer

Q61. Why is bit stuffing required in bit-oriented protocols?

Bit-oriented protocols में bit stuffing क्यों ज़रूरी है?

(A) To increase bandwidth **(B)** To prevent confusion between data and the flag
(C) To reduce collisions **(D)** To detect errors

[Source: BCI Subject Test-9.pdf] · Easy

Q62. The correct order of fields in an Ethernet frame is:

Ethernet frame में fields का सही क्रम है:

(A) Preamble → Destination MAC → Source MAC → Type/Length → Data → FCS **(B)** Destination MAC → Source MAC → FCS → Data → Preamble
(C) Source MAC → Destination MAC → Data → CRC → Type **(D)** Preamble → Source MAC → Destination MAC → Data

[Source: BCI Subject Test-9.pdf] · Easy

Q63. Collision detection and avoidance mechanisms are implemented in:

Collision detection और avoidance mechanisms कहाँ लागू होते हैं?

(A) LLC Sublayer **(B)** Network Layer
(C) MAC Sublayer **(D)** Physical Layer

[Source: BCI Subject Test-9.pdf] · Easy

Q64. Which protocol requires time synchronization among nodes?

किस protocol में nodes के बीच time synchronization चाहिए होती है?

(A) Pure ALOHA **(B)** Slotted ALOHA
(C) CSMA **(D)** CSMA/CD

[Source: BCI Subject Test-9.pdf] · Easy

Q65. The vulnerable period of Pure ALOHA is:

Pure ALOHA का vulnerable period होता है:

(A) T **(B)** 2T
(C) T/2 **(D)** 4T

[Source: BCI Subject Test-9.pdf] · Easy

Q66. Which protocol has the highest efficiency under heavy load?

भारी load पर किस protocol की efficiency सबसे अधिक होती है?

(A) Pure ALOHA **(B)** Slotted ALOHA
(C) CSMA **(D)** CSMA/CD

[Source: BCI Subject Test-9.pdf] · Easy

Q67. Which protocol detects collisions while transmission is in progress?

कौन-सा protocol transmission के दौरान ही collisions detect करता है?

(A) CSMA **(B)** Slotted ALOHA
(C) CSMA/CD **(D)** CSMA/CA

[Source: BCI Subject Test-9.pdf] · Easy

Q68. Which mechanism is used by CSMA/CA to reduce collisions?

CSMA/CA collisions कम करने के लिए कौन-सा mechanism उपयोग करता है?

(A) Collision detection **(B)** Slot restriction
(C) RTS/CTS handshake **(D)** Jam signal

[Source: BCI Subject Test-9.pdf] · Easy

Q69. Bit stuffing refers to:

Bit stuffing का अर्थ है:

(A) Inserting a '0' in the user stream to differentiate it from the flag **(B)** Inserting a '0' in the flag stream to avoid ambiguity
(C) Appending a nibble to the flag sequence **(D)** Appending a nibble to the user data stream

[Source: Unit Test-4 Solution.pdf (OCR), p26] · Medium

Q70. In Ethernet CSMA/CD, the special sequence transmitted by media access management to handle a collision is called:

Ethernet CSMA/CD में collision संभालने के लिए media access management द्वारा भेजा गया विशेष sequence कहलाता है:

(A) Preamble **(B)** Postamble
(C) Jam signal **(D)** None of these

[Source: Unit Test-4 Solution.pdf (OCR), p28] · Medium

Q71. The minimum size of an Ethernet frame (excluding preamble) is:

Ethernet frame का न्यूनतम आकार (preamble को छोड़कर) होता है:

(A) 46 bytes **(B)** 64 bytes
(C) 72 bytes **(D)** 1518 bytes

[Source: BCI Subject Test-9.pdf] · Medium

Q72. The data field size in an Ethernet frame ranges from:

Ethernet frame में data field का आकार रहता है:

(A) 64–1518 bytes **(B)** 46–1500 bytes
(C) 40–1480 bytes **(D)** 32–1500 bytes

[Source: BCI Subject Test-9.pdf] · Medium

Q73. In Ethernet, the minimum frame size is fixed mainly to:

Ethernet में न्यूनतम frame आकार मुख्य रूप से किसलिए तय किया गया है?

(A) Improve throughput **(B)** Reduce overhead
(C) Detect collisions using CSMA/CD **(D)** Increase security

[Source: BCI Subject Test-9.pdf] · Medium

Q74. What is the primary function of the MAC sublayer?

MAC sublayer का मुख्य काम क्या है?

(A) Routing **(B)** Error checking
(C) Addressing **(D)** Channel access

[Source: CN Topic Test-04.pdf] · Medium

Q75. What is the primary role of CSMA/CD in Ethernet?

Ethernet में CSMA/CD की मुख्य भूमिका क्या है?

(A) Error correction **(B)** Collision detection
(C) Address resolution **(D)** Flow control

[Source: CN Topic Test 5.pdf] · Medium

Q76. Why is CSMA/CD not suitable for wireless networks?

CSMA/CD wireless networks के लिए उपयुक्त क्यों नहीं है?

(A) No random access **(B)** Collision detection is not feasible in wireless
(C) High bandwidth requirement **(D)** Requires slot synchronization

[Source: BCI Subject Test-9.pdf] · Medium

Q77. What is multiple access?

Multiple access क्या है?

(A) Accessing a computer from multiple locations **(B)** A kind of network (MAN)
(C) Sharing of one physical link by more than one station **(D)** None of the above

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

Q78. A Token Ring implementation works on the _____ mode of medium access.

Token Ring implementation medium access के _____ mode पर काम करती है.

(A) Contention **(B)** Reservation
(C) Round Robin **(D)** None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p26] · Medium

Q79. CSMA/CD is used in:

CSMA/CD प्रयोग होता है:

(A) Wireless LAN **(B)** Wired Ethernet
(C) Satellite links **(D)** Fiber only / केवल Fiber

[Source: Practice — original (textbook-based)] · Medium

Q80. In which access method does a station wait for a free time slot before transmitting?

किस access method में station भेजने से पहले ख़ाली time slot की प्रतीक्षा करता है?

(A) Pure ALOHA **(B)** Slotted ALOHA
(C) 1-persistent CSMA **(D)** Polling

[Source: Practice — original (textbook-based)] · Medium

Q81. Token passing is an example of which access method?

Token passing किस access method का उदाहरण है?

(A) Random access **(B)** Controlled access
(C) Contention **(D)** Aloha

[Source: Practice — original (textbook-based)] · Medium

Q82. Which field of the Ethernet frame is used for error detection?

Ethernet frame का कौन-सा field error detection के लिए है?

(A) Preamble **(B)** Type field
(C) FCS **(D)** MAC address

[Source: Practice — original (textbook-based)] · Medium

Q83. Gigabit Ethernet over optical fiber is defined by which IEEE standard?

Optical fiber पर Gigabit Ethernet किस IEEE standard में परिभाषित है?

(A) 802.3z **(B)** 802.3ab
(C) 802.3u **(D)** 802.3ae

[Source: Practice — original (textbook-based)] · Medium

Q84. What is the MTU size for Ethernet?

Ethernet के लिए MTU आकार कितना है?

(A) 1024 bytes **(B)** 1500 bytes
(C) 512 bytes **(D)** 2048 bytes

[Source: CN Topic Test 5.pdf] · Hard

Q85. In a slotted ALOHA with G stations attempting to transmit in one slot, the throughput is equal to:

Slotted ALOHA में एक slot में G stations transmit करने का प्रयास करें तो throughput होगा:

(A) $G \times e^{-G}$ **(B)** $2G \times e^{-G}$
(C) e^{-G} **(D)** $G \times e^{-2G}$

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q86. Which of the following Multiple Access Protocols is not suitable for wired networks?

निम्न में से कौन-सा Multiple Access Protocol wired networks के लिए उपयुक्त नहीं है?

(A) CSMA/CD **(B)** TDMA
(C) Token passing **(D)** CSMA/CA

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q87. Network Allocation Vector (NAV) is associated with which of the following MAC technologies?

Network Allocation Vector (NAV) निम्न में से किस MAC technology से जुड़ा है?

(A) Wi-Fi (CSMA/CA) **(B)** CSMA/CD
(C) Bluetooth **(D)** Ethernet

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q88. In Reservation mode of MAC, each station in the network _____ a time slot for a finite or infinite amount of time to access the shared medium.

MAC के Reservation mode में network का हर station shared medium तक पहुँचने के लिए एक time slot _____ करता है.

(A) should reserve **(B)** need not reserve
(C) Can't say **(D)** None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p25] · Hard

Q89. Which one of the following is not used in media access control?

निम्न में से कौन-सा media access control में उपयोग नहीं होता?

(A) Ethernet **(B)** Digital subscriber line (DSL)
(C) FDDI **(D)** None of the mentioned

[Source: CN Topic Test-01.pdf] · Hard

Q90. Which of the following is NOT true with respect to a transparent bridge and a router?

Transparent bridge और router के संबंध में निम्न में से कौन-सा सही नहीं है?

(A) Both bridge and router selectively forward data packets **(B)** A bridge uses IP addresses while a router uses MAC addresses
(C) A bridge builds its table by inspecting incoming packets **(D)** A router can connect between a LAN and a WAN

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q91. Which of the following is NOT a function of the AAL (ATM Adaptation Layer)?

निम्न में से कौन-सा AAL (ATM Adaptation Layer) का काम नहीं है?

(A) Cell header generation **(B)** Handling of lost and misinserted cells
(C) Handling of cell delay variation **(D)** Segmentation and reassembly of user information

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q92. In bit stuffing, after how many consecutive 1-bits is a 0 inserted?

Bit stuffing में कितने लगातार 1-bits के बाद 0 जोड़ा जाता है?

(A) 4 **(B)** 5
(C) 6 **(D)** 8

[Source: Practice — original (textbook-based)] · Hard

Q93. CSMA/CD (collision detection) can be used only when the link operates in:

CSMA/CD (collision detection) तभी प्रयोग हो सकता है जब link चलती हो:

(A) Half-duplex mode **(B)** Full-duplex mode
(C) Simplex mode **(D)** Both half and full duplex / Half और full duplex दोनों

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Hard

Section C — Network Layer: IP, ICMP, ARP & Routing Protocols

Q94. What is the function of ARP?

ARP का काम क्या है?

(A) Resolve IP address to MAC address **(B)** Resolve domain name to IP
(C) Encrypt packets **(D)** Determine the route

[Source: CN Topic Test 6.pdf + Unit Test -2 Solutions.pdf (OCR)] · Medium

Q95. Which tool is used to view the MAC address of a system?

किस system का MAC address देखने के लिए कौन-सा tool उपयोग होता है?

(A) ping **(B)** ipconfig /all
(C) netstat **(D)** arp -a

[Source: BCI_SCI Subject Test-7.pdf] · Medium

Q96. Which protocol is used for path MTU discovery?

Path MTU discovery के लिए कौन-सा protocol उपयोग होता है?

(A) TCP **(B)** UDP
(C) ICMP **(D)** ARP

[Source: BCI_SCI Subject Test-7.pdf] · Medium

Q97. What is the function of IGMP in networking?

Networking में IGMP का काम क्या है?

(A) Manage IP address allocation **(B)** Route datagrams
(C) Manage multicast group membership **(D)** Handle device discovery

[Source: CN Topic Test 5.pdf] · Medium

Q98. What is the role of the TTL field in an IP packet?

IP packet में TTL field की भूमिका क्या है?

(A) Authentication **(B)** Packet size control
(C) Packet lifespan control **(D)** Encryption

[Source: CN Topic Test 6.pdf] · Medium

Q99. Which field in the IPv4 header indicates the priority of the packet?

IPv4 header में कौन-सा field packet की प्राथमिकता बताता है?

(A) Protocol **(B)** TTL
(C) Type of Service (ToS) **(D)** Flags

[Source: CN Topic Test 6.pdf] · Medium

Q100. Which of the following uses a logical addressing scheme?

निम्न में से कौन-सा logical addressing scheme उपयोग करता है?

(A) MAC address **(B)** IP address
(C) ARP **(D)** DNS

[Source: CN Topic Test 5.pdf] · Medium

Q101. Which algorithm is used by OSPF to compute the shortest path?

OSPF shortest path निकालने के लिए कौन-सा algorithm उपयोग करता है?

(A) Bellman-Ford **(B)** Dijkstra
(C) A* **(D)** Kruskal

[Source: CN Topic Test 6.pdf] · Medium

Q102. Which one is a distance-vector routing protocol?

निम्न में से कौन-सा distance-vector routing protocol है?

(A) RIP **(B)** OSPF
(C) BGP **(D)** IS-IS

[Source: CN Topic Test 6.pdf] · Medium

Q103. What is a major drawback of distance vector routing?

Distance vector routing की एक बड़ी कमी क्या है?

(A) High complexity **(B)** Count-to-infinity problem
(C) Slow convergence **(D)** Lack of security

[Source: CN Topic Test 5.pdf] · Medium

Q104. BGP is mainly used for which type of routing, and it belongs to which algorithm family?

BGP मुख्य रूप से किस प्रकार के routing के लिए उपयोग होता है और यह किस algorithm परिवार का है?

(A) Intra-AS routing; link-state **(B)** Inter-AS routing; path-vector (distance-vector family)
(C) Static routing; flooding **(D)** Host routing; Dijkstra

[Source: CN Topic Test 5.pdf + CN Topic Test 6.pdf] · Medium

Q105. In the early ARPANET, which routing method was used?

प्रारंभिक ARPANET में कौन-सी routing method उपयोग होती थी?

(A) Distance vector routing **(B)** Link state routing
(C) Least cost routing **(D)** Flooding

[Source: Unit Test-4 Solution.pdf (OCR), p8] · Medium

Q106. Which of these protocols allows loop-free switching in Layer 2 networks?

निम्न में से कौन-सा protocol Layer 2 networks में loop-free switching देता है?

(A) RIP **(B)** OSPF
(C) STP (Spanning Tree Protocol) **(D)** EIGRP

[Source: CN Topic Test 6.pdf] · Medium

Q107. MPLS operates between which two layers?

MPLS किन दो layers के बीच काम करता है?

(A) Layer 1 and 2 **(B)** Layer 2 and 3
(C) Layer 3 and 4 **(D)** Layer 4 and 7

[Source: CN Topic Test 5.pdf] · Medium

Q108. What does the 'traceroute' command do?

'traceroute' command क्या करता है?

(A) Lists all devices on a network **(B)** Tracks the path a packet takes to the destination
(C) Resolves domain names **(D)** Opens ports on the firewall

[Source: CN Topic Test 5.pdf] · Medium

Q109. Which protocol does traceroute use on UNIX/Linux systems?

UNIX/Linux systems पर traceroute कौन-सा protocol उपयोग करता है?

(A) TCP **(B)** UDP
(C) ICMP Echo **(D)** IGMP

[Source: CN Topic Test 5.pdf] · Medium

Q110. Which command is used to see the routing table in Linux?

Linux में routing table देखने के लिए कौन-सा command उपयोग होती है?

(A) ifconfig **(B)** netstat -r
(C) ping **(D)** ipconfig

[Source: CN Topic Test 5.pdf] · Medium

Q111. Which protocol is used for dynamic IP address assignment?

Dynamic IP address assignment के लिए कौन-सा protocol उपयोग होता है?

(A) DNS **(B)** ARP
(C) DHCP **(D)** FTP

[Source: CN Topic Test-03.pdf + CN Topic Test 6.pdf] · Medium

Q112. Why is the TTL field decremented at each router?

हर router पर TTL field क्यों घटाया जाता है?

(A) To measure speed / Speed नापने के लिए **(B)** To prevent infinite looping of packets / Packets के अनंत चक्कर रोकने के लिए
(C) To encrypt the packet / Packet encrypt करने के लिए **(D)** To compress the packet / Packet compress करने के लिए

[Source: Practice — original (textbook-based)] · Medium

Q113. Which routing protocol uses hop count as its metric with a maximum of 15?

कौन-सा routing protocol hop count metric रखता है (अधिकतम 15)?

(A) OSPF **(B)** BGP
(C) RIP **(D)** ICMP

[Source: Practice — original (textbook-based)] · Medium

Q114. Ping utility uses which ICMP messages?

Ping utility किस ICMP messages का प्रयोग करती है?

(A) Redirect **(B)** Echo request and echo reply / Echo request और echo reply
(C) Time exceeded **(D)** Address mask

[Source: Practice — original (textbook-based)] · Medium

Q115. BGP is used for:

BGP प्रयोग होता है:

(A) Routing inside one LAN / एक LAN के अंदर routing के लिए **(B)** Routing between autonomous systems / Autonomous systems के बीच routing के लिए
(C) Error reporting / Error reporting के लिए **(D)** Address resolution / Address resolution के लिए

[Source: Practice — original (textbook-based)] · Medium

Q116. Congestion control in the network layer is mainly concerned with:

Network layer का congestion control मुख्यतः संबंधित है:

(A) Too many packets in the network / Network में बहुत अधिक packets से **(B)** Slow receiver / धीमे receiver से
(C) Broken cables / टूटी cables से **(D)** Wrong addresses / ग़लत addresses से

[Source: Practice — original (textbook-based)] · Medium

Q117. OSPF exchanges routing information directly over IP using IP protocol number:

OSPF routing information IP पर सीधे भेजता है — IP protocol number होता है:

(A) 89 **(B)** 179
(C) 520 **(D)** 6

[Source: Practice — original (textbook-based)] · Medium

Q118. BGP, the protocol that routes between autonomous systems on the Internet, uses:

BGP — जो Internet पर autonomous systems के बीच routing करता है — प्रयोग करता है:

(A) TCP port 179 **(B)** IP protocol 89
(C) UDP port 520 **(D)** TCP port 23

[Source: Practice — original (textbook-based)] · Medium

Q119. The RIP routing protocol uses which transport protocol and port number?

RIP routing protocol कौन-सा transport protocol और port number प्रयोग करता है?

(A) UDP port 520 **(B)** TCP port 520
(C) UDP port 161 **(D)** TCP port 89

[Source: Practice — original (textbook-based)] · Medium

Q120. If the Protocol field of an IPv4 header is 17, the transport-layer protocol carried in the datagram is:

यदि IPv4 header का Protocol field 17 हो, तो datagram में लिया गया transport-layer protocol है:

(A) UDP **(B)** TCP
(C) ICMP **(D)** IGMP

[Source: Practice — original (textbook-based)] · Medium

Q121. RSVP (Resource ReSerVation Protocol) is used by hosts to:

RSVP (Resource ReSerVation Protocol) का उपयोग hosts करते हैं:

(A) Request reservation of network resources (bandwidth) along a path for QoS flows / QoS flows के लिए रास्ते में network संसाधन (bandwidth) आरक्षित करने के अनुरोध हेतु
(B) Assign IP addresses automatically / स्वतः IP addresses देने के लिए
(C) Resolve domain names / Domain names हल करने के लिए
(D) Exchange routing tables between routers / Routers के बीच routing tables बाँटने के लिए

[Source: Practice — original (textbook-based)] · Medium

Q122. ICMP is a ..... layer protocol.

ICMP एक ..... layer protocol है.

(A) Data link **(B)** Transport
(C) Application **(D)** Network

[Source: RPSC Programmer 2024 Paper-1 (with official key) + Random Test- Rajasthan PYQ.pdf (OCR)] · Hard

Q123. Which of the following is NOT a valid ICMP error message?

निम्न में से कौन-सा वैध ICMP error message नहीं है?

(A) Destination Unreachable **(B)** Time Exceeded
(C) Echo Request **(D)** Port Unreachable

[Source: CN Topic Test 5.pdf] · Hard

Q124. Which field in the IP header helps in fragmentation?

IP header में कौन-सा field fragmentation में मदद करता है?

(A) TTL **(B)** Header Length
(C) Identification **(D)** Protocol

[Source: CN Topic Test 6.pdf] · Hard

Q125. How many bits are there in the 'Fragment Offset' field of the IPv4 header?

IPv4 header के 'Fragment Offset' field में कितने bits होते हैं?

(A) 16 bits **(B)** 10 bits
(C) 13 bits **(D)** 32 bits

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q126. In IPv4, packet fragmentation and reassembly is handled by:

IPv4 में packet fragmentation और reassembly कैसे होता है?

(A) the More Fragments flag and Fragment Offset field
(B) the MAC Header
(C) the Header Checksum **(D)** encapsulating packets in larger frames

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q127. What is fragmentation in IP networking?

IP networking में fragmentation क्या है?

(A) Splitting data into packets **(B)** Splitting large packets so they fit a smaller MTU link
(C) Combining multiple packets **(D)** Encrypting packets

[Source: CN Topic Test 6.pdf] · Hard

Q128. The size of the 'total length field' of an IPv4 datagram is:

IPv4 datagram के 'total length field' का आकार है:

(A) 16 bits **(B)** 32 bits
(C) 8 bits **(D)** 4 bits

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q129. What is the minimum header size of an IPv4 packet?

IPv4 packet का न्यूनतम header आकार कितना है?

(A) 16 bytes **(B)** 10 bytes
(C) 20 bytes **(D)** 32 bytes

[Source: Networking MCQ bank (MQuestions)] · Hard

Q130. In the IPv6 header, the traffic class field is similar to which field in the IPv4 header?

IPv6 header में traffic class field, IPv4 header के किस field जैसा है?

(A) Fragmentation field **(B)** Flow label
(C) Type of Service (ToS) field **(D)** Options field

[Source: Sanfoundry CN MCQs — IPv6] · Hard

Q131. How many maximum hops are allowed in RIP?

RIP में अधिकतम कितने hops की अनुमति है?

(A) 8 **(B)** 16
(C) 15 **(D)** 32

[Source: CN Topic Test 6.pdf] · Hard

Q132. How does RIP prevent routing loops? (A) Poison reverse (B) Sequence numbers (C) Split horizon

RIP routing loops कैसे रोकता है? (A) Poison reverse (B) Sequence numbers (C) Split horizon

(A) A only **(B)** C only
(C) B only **(D)** Both A and C

[Source: CN Topic Test 6.pdf] · Hard

Q133. Dynamic routing protocols enable routers to:

Dynamic routing protocols routers को सक्षम बनाते हैं:

(A) Dynamically discover and maintain routes **(B)** Distribute routing updates to other routers
(C) Reach agreement with other routers about network topology **(D)** All of the above

[Source: Unit Test-4 Solution.pdf (OCR), p27] · Hard

Q134. Which of the following protocols may be used as zeroconf and plug-and-play?

निम्न में से कौन-सा protocol zeroconf और plug-and-play के रूप में उपयोग हो सकता है?

(A) TCP **(B)** DHCP
(C) NAT **(D)** ICMP

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q135. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. The probability that exactly one station transmits in a slot is:

एक slotted LAN में n stations हैं. हर time slot में हर station p प्रायिकता से transmit करने का प्रयास करता है. किसी slot में ठीक एक station transmit करने की प्रायिकता है:

(A) $n \times p \times (1-p)^{n-1}$ **(B)** $p \times (1-p)^{n-1}$
(C) $1-(1-p)^{n-1}$ **(D)** $(1-p)^{n-1}$

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q136. One of the header fields in an IP datagram is the Time to Live (TTL). Which statement best explains the need for this field?

IP datagram के header fields में से एक Time to Live (TTL) है. निम्न में से कौन-सा कथन इस field की आवश्यकता सबसे अच्छे से बताता है?

(A) It can be used to prioritize packets **(B)** It can be used to reduce delays
(C) It can be used to optimize throughput **(D)** It can be used to prevent packet looping

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q137. Suppose the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Acknowledgement transmission time and processing times are negligible. The minimum frame size in bytes to achieve a link utilisation of at least 50% is _____.

64 kbps की link पर stop-and-wait protocol है, propagation delay 20 ms है (ACK transmission व processing नगण्य)। कम से कम 50% link utilisation के लिए न्यूनतम frame size (bytes में) कितनी हो?

(A) 160 **(B)** 320
(C) 640 **(D)** 1280

[Source: GATE CS 2015 Set-1] · Hard

Q138. A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Acknowledgements have negligible transmission delay and the same propagation delay as data; processing delays are negligible. The efficiency of the stop-and-wait protocol is exactly 25%. The one-way propagation delay (in milliseconds) is _____.

एक link की speed 10^6 bits/sec है और packets 1000 bytes के हैं। ACK का transmission delay नगण्य, propagation delay data के बराबर है। Stop-and-wait की efficiency ठीक 25% है। एक-तरफ़ा propagation delay (ms में) कितना है?

(A) 8 **(B)** 10
(C) 12 **(D)** 20

[Source: GATE CS 2015 Set-2] · Hard

Q139. Consider the following statements regarding the slow start phase of TCP congestion control (cwnd = congestion window, MSS = maximum segment size): (i) cwnd increases by 2 MSS on every successful acknowledgement. (ii) cwnd approximately doubles on every successful acknowledgement. (iii) cwnd increases by 1 MSS every round trip time. (iv) cwnd approximately doubles every round trip time. Which one of the following is correct?

TCP congestion control के slow start चरण (cwnd = congestion window, MSS = maximum segment size) के बारे में विचार कीजिए: (i) हर successful acknowledgement पर cwnd 2 MSS बढ़ता है। (ii) हर successful acknowledgement पर cwnd लगभग दोगुना होता है। (iii) हर round trip time में cwnd 1 MSS बढ़ता है। (iv) हर round trip time में cwnd लगभग दोगुना होता है। क्या सही है?

(A) (i) and (iii) only / केवल (i) और (iii) **(B)** (ii) and (iv) only / केवल (ii) और (iv)
(C) (iv) only / केवल (iv) **(D)** (ii) only / केवल (ii)

[Source: GATE CS 2018] · Hard

Q140. An IP datagram of 1000 bytes (20-byte header) is fragmented on a link with MTU 200 bytes. The number of data bytes in the FIRST fragment is:

1000 bytes का datagram (20-byte header), MTU 200 link पर टूटा। पहले fragment में data bytes होंगे:

(A) 180 **(B)** 200
(C) 980 **(D)** 160

[Source: Practice — original (textbook-based)] · Hard

Q141. A router receives a datagram whose TTL becomes 0 after it decrements the field. The router will:

एक router को datagram मिला जिसका TTL घटाने पर 0 हो गया। router क्या करेगा?

(A) Discard the datagram and send an ICMP Time Exceeded message to the source / Datagram नष्ट करके source को ICMP Time Exceeded भेजेगा **(B)** Forward it with TTL 0 / TTL 0 के साथ आगे भेजेगा
(C) Send it back to the previous router / पिछले router को वापस भेजेगा **(D)** Change TTL to 255 and forward / TTL 255 करके आगे भेजेगा

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q142. In OSPF, on a multiaccess network the Designated Router (DR) is the router with:

OSPF में multiaccess network पर Designated Router (DR) होता है वह router जिसके पास:

(A) The highest OSPF interface priority (tie → highest router ID) / सबसे अधिक OSPF interface priority (बराबरी पर → सबसे बड़ा router ID) **(B)** The lowest IP address / सबसे कम IP address
(C) The highest bandwidth link / सबसे अधिक bandwidth वाला link **(D)** The oldest OSPF process / सबसे पुराना OSPF process

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q143. Two OSPF routers are stuck in the EXSTART/EXCHANGE neighbour state. The most likely cause is:

दो OSPF routers EXSTART/EXCHANGE neighbour state में अटके हैं। सबसे संभावित कारण है:

(A) An MTU mismatch on the connecting interfaces / जोड़ने वाले interfaces पर MTU का अंतर **(B)** Different OSPF process IDs / भिन्न OSPF process IDs
(C) Different hostnames / भिन्न hostnames **(D)** Different IOS/OS versions / भिन्न IOS/OS versions

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q144. The administrative distance of a static route, an EIGRP route and an OSPF route are respectively:

Static route, EIGRP route और OSPF route का administrative distance क्रमशः होता है:

(A) 1, 90 and 110 / 1, 90 और 110 **(B)** 0, 110 and 120 / 0, 110 और 120
(C) 1, 110 and 120 / 1, 110 और 120 **(D)** 10, 90 and 100 / 10, 90 और 100

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q145. In IPv4 fragmentation, the LAST fragment has the More Fragments (MF) flag set to:

IPv4 fragmentation में, अंतिम fragment का More Fragments (MF) flag होता है:

(A) 0 **(B)** 1
(C) Its fragment offset value / उसका fragment offset मान
(D) The same as the first fragment / पहले fragment के बराबर

[Source: Practice — original (textbook-based)] · Hard

Q146. The Fragment Offset field in the IPv4 header is used to:

IPv4 header का Fragment Offset field प्रयोग होता है:

(A) Place the fragment at its correct position in the original datagram during reassembly / Reassembly के दौरान fragment को मूल datagram में सही स्थान पर रखने के लिए **(B)** Count the total number of fragments / कुल fragments गिनने के लिए
(C) Encrypt fragment payloads / Fragment payloads encrypt करने के लिए **(D)** Set the fragment's lifetime / Fragment का जीवन-काल तय करने के लिए

[Source: Practice — original (textbook-based)] · Hard

Q147. The data field of an IPv4 datagram can carry:

IPv4 datagram का data field ले जा सकता है:

(A) TCP segment, UDP datagram, or ICMP message — all of these / TCP segment, UDP datagram, या ICMP message — ये सब **(B)** Only TCP segments / सिर्फ TCP segments
(C) Only UDP datagrams / सिर्फ UDP datagrams **(D)** Only application data / सिर्फ application data

[Source: Practice — original (textbook-based)] · Hard

Section D — Transport Layer: TCP/UDP, Ports & Sockets

Q148. A client opens a TCP connection to a web server on port 80 using source port 49152. What is the complete socket pair?

एक client source port 49152 से port 80 वाले web server को TCP connection खोलता है. पूरा socket pair क्या होगा?

(A) 192.168.1.10:49152 ↔ 172.16.0.5:80 **(B)** 192.168.1.10:80 ↔ 172.16.0.5:49152
(C) 49152 ↔ 80 **(D)** 80 ↔ 49152

[Source: BCI Subject Test-8.pdf] · Easy

Q149. Which port is used by a DHCP client when requesting an IP address?

IP address माँगते समय DHCP client कौन-सा port उपयोग करता है?

(A) UDP 67 **(B)** UDP 68
(C) TCP 68 **(D)** TCP 67

[Source: BCI Subject Test-8.pdf] · Easy

Q150. Which of the following is a connectionless transport service?

निम्न में से कौन-सी connectionless transport service है?

(A) TCP **(B)** ARP
(C) UDP **(D)** FTP

[Source: BCI_SCI Subject Test-7.pdf + Oliveboard SSC CGL bank] · Medium

Q151. What is a socket in networking?

Networking में socket क्या है?

(A) IP address + MAC address **(B)** IP address + Port number
(C) MAC address + Port number **(D)** IP address + DNS name

[Source: CN Topic Test 6.pdf] · Medium

Q152. What term is used for an endpoint in an inter-process communication across a network?

Network पर inter-process communication में endpoint के लिए कौन-सा term उपयोग होता है?

(A) Port **(B)** Machine
(C) Socket **(D)** Pipe

[Source: CN Topic Test-02.pdf] · Medium

Q153. What is the purpose of the SYN flag in TCP?

TCP में SYN flag का उद्देश्य क्या है?

(A) To end a session **(B)** To initiate (open) a connection
(C) To reset a connection **(D)** For data transmission

[Source: CN Topic Test 6.pdf] · Medium

Q154. In which scenario will TCP's three-way handshake fail?

किस स्थिति में TCP का three-way handshake विफल होगा?

(A) Firewall blocks the SYN-ACK **(B)** High latency
(C) ACK is slightly delayed **(D)** Server is busy

[Source: CN Topic Test 5.pdf] · Medium

Q155. In TCP, which state follows the SYN-SENT state in the normal case?

TCP में सामान्य स्थिति में SYN-SENT के बाद कौन-सी state आती है?

(A) SYN-RECEIVED **(B)** LISTEN
(C) ESTABLISHED **(D)** CLOSED

[Source: CN Topic Test 6.pdf] · Medium

Q156. Which one of the following attacks exploits TCP's three-way handshake?

निम्न में से कौन-सा attack TCP के three-way handshake का फ़ायदा उठाती है?

(A) SYN Flood **(B)** DNS Spoofing
(C) Man in the Middle **(D)** ARP Poisoning

[Source: CN Topic Test 5.pdf] · Medium

Q157. TCP initially uses which congestion control technique when a connection starts?

Connection शुरू होने पर TCP शुरू में कौन-सी congestion control technique उपयोग करता है?

(A) Congestion avoidance **(B)** Fast recovery
(C) Slow start **(D)** Leaky bucket

[Source: CN Topic Test 6.pdf] · Medium

Q158. What is the purpose of TCP's congestion window (cwnd)?

TCP के congestion window (cwnd) का उद्देश्य क्या है?

(A) Track unacknowledged packets **(B)** Control the sending rate to avoid network congestion
(C) Handle encryption **(D)** Prioritize packets

[Source: CN Topic Test 5.pdf] · Medium

Q159. What kind of delay does TCP's window size help handle on a long, high-bandwidth path?

लंबी, high-bandwidth path पर TCP का window size किस प्रकार की देर संभालने में मदद करता है?

(A) Transmission delay **(B)** Queuing delay
(C) Propagation delay **(D)** Bandwidth-delay product

[Source: CN Topic Test 5.pdf] · Medium

Q160. In which protocol is 'port knocking' used as a security mechanism?

किस protocol में 'port knocking' security mechanism के रूप में उपयोग होता है?

(A) TCP **(B)** UDP
(C) SSH **(D)** IP

[Source: CN Topic Test 6.pdf] · Medium

Q161. Which of the following supports QoS (Quality of Service) through label switching?

निम्न में से कौन-सा label switching के ज़रिए QoS (Quality of Service) देता है?

(A) TCP **(B)** UDP
(C) MPLS **(D)** ARP

[Source: CN Topic Test 6.pdf] · Medium

Q162. Virtual circuits are associated with which type of service?

Virtual circuits किस प्रकार की service से जुड़े हैं?

(A) Connectionless service **(B)** Error-free service
(C) Segmentation **(D)** Connection-oriented service

[Source: CN Topic Test-02.pdf] · Medium

Q163. Transport layer multiplexing allows:

Transport layer multiplexing की अनुमति देता है:

(A) Multiple hosts to share one IP **(B)** Multiple processes to share one network connection
(C) Multiple routers to share a routing table **(D)** Domain name resolution

[Source: BCI_SCI Subject Test-7.pdf] · Medium

Q164. Which protocol must use both TCP and UDP for different operations?

कौन-सा protocol अलग-अलग कामों के लिए TCP और UDP दोनों उपयोग करता है?

(A) DNS **(B)** HTTP
(C) POP3 **(D)** Telnet

[Source: BCI Subject Test-8.pdf + CN Topic Test 5.pdf] · Medium

Q165. Which transport-layer protocol is used to support electronic mail delivery between machines?

Machines के बीच electronic mail delivery के लिए कौन-सा transport-layer protocol उपयोग होता है?

(A) SNMP **(B)** IP
(C) SMTP **(D)** TCP

[Source: CN Topic Test-03.pdf] · Medium

Q166. During TCP connection termination, after receiving FIN the receiver sends:

TCP connection समाप्ति में FIN पाने के बाद receiver भेजता है:

(A) SYN **(B)** ACK
(C) RST **(D)** PSH

[Source: Practice — original (textbook-based)] · Medium

Q167. In TCP slow start, the congestion window size:

TCP slow start में congestion window size:

(A) Increases by 1 MSS per RTT / प्रति RTT 1 MSS बढ़ती है
(B) Doubles approximately every RTT / हर RTT में लगभग दोगुनी होती है
(C) Remains constant / स्थिर रहती है **(D)** Decreases every ACK / हर ACK पर घटती है

[Source: Practice — original (textbook-based)] · Medium

Q168. Which of the following uses UDP?

निम्न में से कौन UDP प्रयोग करता है?

(A) FTP **(B)** SMTP
(C) DNS query **(D)** SSH

[Source: Practice — original (textbook-based)] · Medium

Q169. After a TCP connection is closed, the side performing the active close waits in the TIME-WAIT state for:

TCP connection बंद होने के बाद active close करने वाला पक्ष TIME-WAIT state में रहता है:

(A) 2 × MSL (Maximum Segment Lifetime) **(B)** 1 × RTT
(C) 3 × RTT **(D)** One congestion timeout / एक congestion timeout

[Source: Practice — original (textbook-based)] · Medium

Q170. How many bits are used for port addressing at the transport layer?

Transport layer पर port addressing के लिए कितने bits उपयोग होते हैं?

(A) 8 **(B)** 10
(C) 16 **(D)** 32

[Source: BCI_SCI Subject Test-7.pdf] · Hard

Q171. The three-way handshake mechanism in TCP aims to:

TCP का three-way handshake mechanism किस उद्देश्य से होता है?

(A) Secure the communication **(B)** Send bits from one device to another
(C) Perform error detection and correction **(D)** Establish a connection and synchronize sequence numbers

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q172. What type of attack is prevented by using random initial sequence numbers in TCP?

TCP में random initial sequence numbers उपयोग करने से किस प्रकार की attack रुकती है?

(A) Smurf attack **(B)** SYN Flooding
(C) TCP session hijacking **(D)** Ping of Death

[Source: CN Topic Test 5.pdf] · Hard

Q173. Which field in the TCP header is used for flow control?

TCP header में flow control के लिए कौन-सा field उपयोग होता है?

(A) Window Size **(B)** Sequence Number
(C) Acknowledgment Number **(D)** Checksum

[Source: CN Topic Test 6.pdf] · Hard

Q174. Which TCP header field ensures that data arrives in order?

TCP header का कौन-सा field सुनिश्चित करता है कि data क्रम में पहुँचे?

(A) Checksum **(B)** Flags
(C) Sequence Number **(D)** Port Number

[Source: CN Topic Test 5.pdf] · Hard

Q175. Which field in the TCP header ensures data integrity?

TCP header का कौन-सा field data integrity सुनिश्चित करता है?

(A) Urgent Pointer **(B)** Sequence Number
(C) Checksum **(D)** Acknowledgment Number

[Source: CN Topic Test 5.pdf] · Hard

Q176. Which of the following is NOT a property of TCP?

निम्न में से कौन-सा TCP का गुण नहीं है?

(A) Reliable **(B)** Ordered delivery
(C) Connection-oriented **(D)** Broadcast

[Source: CN Topic Test 5.pdf] · Hard

Q177. Which of the following provides reliable, connection-oriented communication?

निम्न में से कौन-सा भरोसेमंद, connection-oriented communication देता है?

(A) TCP **(B)** IP
(C) UDP **(D)** All of the above

[Source: MQuestions DCN bank + Oliveboard SSC CGL bank] · Hard

Q178. Which of the following statements about TCP/IP is INCORRECT?

TCP/IP के बारे में निम्न में से कौन-सा कथन गलत है?

(A) TCP/IP has high latency **(B)** Data packets sent over TCP/IP are private
(C) TCP is used by Secure Shell (SSH) **(D)** TCP/IP uses checksums to check errors

[Source: DSSSB TGT CS (via Testbook solved)] · Hard

Q179. Why are port numbers included in the TCP header of a segment?

Segment के TCP header में port numbers क्यों रखे जाते हैं?

(A) To indicate the correct router interface for forwarding **(B)** To identify which switch ports should receive the frame
(C) To determine which Layer 3 protocol encapsulates the data **(D)** To enable a receiving host to forward the data to the appropriate application

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q180. Which one of the following fields of an IP header is NOT modified by a typical IP router?

IP header का कौन-सा field एक सामान्य IP router द्वारा संशोधित नहीं किया जाता?

(A) Checksum **(B)** Source address
(C) Time to Live (TTL) **(D)** Length

[Source: GATE CS 2015 Set-1] · Hard

Q181. Which of the following fields of an IP header is/are ALWAYS modified by any router before it forwards the IP packet?

IP packet forward करने से पहले हर router IP header के कौन-से fields हमेशा बदलता है?

(A) TTL only / केवल TTL **(B)** Checksum only / केवल Checksum
(C) TTL and header checksum / TTL और header checksum **(D)** Source address and TTL / Source address और TTL

[Source: GATE CS 2024 Set-2] · Hard

Q182. Which of the following fields is/are modified in the IP header of a packet going out of a Network Address Translation (NAT) device from an internal network to an external network?

NAT device से internal network से external network की तरफ़ जाते packet के IP header में कौन-से fields बदले जाते हैं?

(A) Source IP address only / केवल Source IP address **(B)** Destination IP address only / केवल Destination IP address
(C) Source IP address and header checksum / Source IP address और header checksum **(D)** TTL and destination address / TTL और destination address

[Source: GATE CS 2024 Set-1] · Hard

Q183. Consider sending an IP datagram of size 1420 bytes (including a 20-byte IP header) over a path of two links with a router between them. The first link has MTU 542 bytes and the second link has MTU 360 bytes. The number of fragments delivered at the receiver is _____.

1420 bytes का IP datagram (20-byte header सहित) दो links के रास्ते भेजा जा रहा है, बीच में एक router है। पहले link का MTU 542 bytes और दूसरे का 360 bytes है। Receiver पर कुल कितने fragments पहुँचेंगे?

(A) 4 **(B)** 5
(C) 6 **(D)** 8

[Source: GATE CS 2024 Set-1] · Hard

Q184. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, and the sequence numbers of the first and the last bytes of the payload, respectively, are:

एक IPv4 datagram में M bit = 0, HLEN = 10, total length = 400 और fragment offset = 300 है। इस datagram की स्थिति तथा payload के पहले और आख़िरी bytes के sequence numbers क्रमशः हैं:

(A) Last fragment, 2400 and 2759 / Last fragment, 2400 और 2759 **(B)** First fragment, 2400 and 2759 / First fragment, 2400 और 2759
(C) Last fragment, 300 and 659 / Last fragment, 300 और 659 **(D)** Last fragment, 2400 and 2779 / Last fragment, 2400 और 2779

[Source: GATE CS 2013] · Hard

Q185. Consider the following three statements about link state and distance vector routing protocols for a large network with 500 nodes and 4000 links: S1: The computational overhead in link state protocols is higher than in distance vector protocols. S2: A distance vector protocol (with split horizon) avoids persistent routing loops, but a link state protocol does not. S3: After a topology change, a link state protocol converges faster than a distance vector protocol. Which one is correct?

500 nodes और 4000 links वाले बड़े network के लिए link state व distance vector routing protocols के बारे में तीन कथन: S1: Link state में computational overhead distance vector से अधिक है। S2: Distance vector (split horizon सहित) persistent loops से बचता है, पर link state नहीं। S3: Topology बदलने पर link state, distance vector से तेज़ converge करता है। कौन-सा सही है?

(A) S1, S2 and S3 are all true / S1, S2, S3 तीनों सही **(B)** S1, S2 and S3 are all false / S1, S2, S3 तीनों गलत
(C) S1 and S2 are true, S3 is false / S1 और S2 सही, S3 गलत
(D) S1 and S3 are true, S2 is false / S1 और S3 सही, S2 गलत

[Source: GATE CS 2014 Set-1] · Hard

Q186. Which one of the following is TRUE about the interior gateway routing protocols RIP (Routing Information Protocol) and OSPF (Open Shortest Path First)?

Interior gateway routing protocols RIP (Routing Information Protocol) और OSPF (Open Shortest Path First) के बारे में कौन-सा कथन सही है?

(A) RIP uses distance vector routing and OSPF uses link state routing / RIP distance vector और OSPF link state routing प्रयोग करता है **(B)** OSPF uses distance vector routing and RIP uses link state routing / OSPF distance vector और RIP link state routing प्रयोग करता है
(C) Both RIP and OSPF use link state routing / दोनों link state routing प्रयोग करते हैं **(D)** Both RIP and OSPF use distance vector routing / दोनों distance vector routing प्रयोग करते हैं

[Source: GATE CS 2014 Set-2] · Hard

Q187. Consider the following statements about the functionality of an IP based router: I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. Which of the above statement(s) is/are TRUE?

IP आधारित router की कार्यप्रणाली के बारे में विचार: I. Router forwarding के दौरान IP packets को संशोधित नहीं करता। II. Router के लिए कोई routing protocol लागू करना आवश्यक नहीं है। III. यदि outgoing link का MTU आने वाले packet से बड़ा हो तो router को fragments जोड़ने चाहिए। इनमें कौन-सा कथन सही है?

(A) I only / केवल I **(B)** II only / केवल II
(C) I and III only / केवल I और III **(D)** None of the above / कोई नहीं

[Source: GATE CS 2020] · Hard

Q188. Host A (on TCP/IPv4 network A) sends an IP datagram D to host B (on TCP/IPv4 network B) with no transmission error. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D? I. TTL II. Checksum III. Fragment Offset

Host A ने (TCP/IPv4 network A से) बिना किसी त्रुटि के IP datagram D, host B को भेजा। D के B पर पहुँचने पर निम्न में से कौन-से IP header fields मूल datagram से भिन्न हो सकते हैं? I. TTL II. Checksum III. Fragment Offset

(A) I only / केवल I **(B)** II and III only / केवल II और III
(C) I and II only / केवल I और II **(D)** I, II and III / I, II और III तीनों

[Source: GATE CS 2014 Set-3] · Hard

Q189. The size of the TCP header (without options) is:

TCP header का size (options बिना) होता है:

(A) 8 bytes **(B)** 20 bytes
(C) 24 bytes **(D)** 60 bytes

[Source: Practice — original (textbook-based)] · Hard

Q190. The size of the UDP header is:

UDP header का size होता है:

(A) 8 bytes **(B)** 12 bytes
(C) 20 bytes **(D)** 64 bytes

[Source: Practice — original (textbook-based)] · Hard

Q191. TCP is called a byte-stream protocol because:

TCP को byte-stream protocol कहते हैं क्योंकि:

(A) It sends only bytes / यह सिर्फ़ bytes भेजता है **(B)** It does not preserve message boundaries / यह message सीमाएँ नहीं रखता
(C) It cannot send text / यह text नहीं भेज सकता **(D)** It works only with bytes files / यह सिर्फ़ byte files के साथ चलता है

[Source: Practice — original (textbook-based)] · Hard

Q192. The 8-byte UDP header consists of which fields?

8-byte UDP header में कौन-से fields होते हैं?

(A) Source port, destination port, length, checksum
(B) Source port, destination port, sequence number, ack number
(C) Source port, destination port, window, checksum
(D) Ports, length, checksum, urgent pointer

[Source: Practice — original (textbook-based)] · Hard

Q193. TCP provides a byte-stream service. This means that:

TCP byte-stream service देता है — इसका अर्थ है कि:

(A) Message boundaries are NOT preserved — a record may be split across segments or two records merged into one / Message boundaries नहीं बनतीं — एक record कई segments में बँट सकता है या दो records एक में मिल सकते हैं **(B)** Each send() call arrives as one segment / हर send() एक segment के रूप में पहुँचता है
(C) Bytes arrive in fixed 512-byte blocks / Bytes 512-byte के तय blocks में आते हैं **(D)** The receiver delivers bytes in the order they were received, even if reordered on the way / Receiver bytes उसी क्रम में देता है जिसमें प्राप्त हुए, भले रास्ते में बदले हों

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q194. The TCP Maximum Segment Size (MSS) is typically the path MTU minus _____ bytes of headers:

TCP Maximum Segment Size (MSS) सामान्यतः path MTU में से _____ bytes के headers घटाने पर मिलती है:

(A) 40 **(B)** 20
(C) 64 **(D)** 54

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Section E — Session, Presentation & Application Layer

Q195. A server receives a packet with destination port 443. This implies:

एक server को destination port 443 वाला packet मिलता है. इसका अर्थ है:

(A) HTTP request **(B)** HTTPS traffic
(C) FTP command **(D)** Email traffic

[Source: BCI Subject Test-8.pdf] · Easy

Q196. For what application is the HTTP protocol used?

HTTP protocol किस काम के लिए उपयोग होता है?

(A) To increase upload speed **(B)** To develop a website
(C) To deliver data on the World Wide Web **(D)** To download images only

[Source: SSC MTS computer PYQ (via Testbook solved)] · Easy

Q197. The structure or format of data is called:

Data का structure या format कहलाता है:

(A) Syntax **(B)** Struct
(C) Semantic **(D)** None of the above

[Source: Unit Test-4 Solution.pdf (OCR), p14] · Medium

Q198. Which protocol layer uses protocols such as WWW, HTTP, FTP, SMTP and e-mail?

कौन-सी protocol layer WWW, HTTP, FTP, SMTP और e-mail जैसे protocols उपयोग करती है?

(A) Application layer **(B)** Transport layer
(C) Internet layer **(D)** Hardware layer

[Source: CN Topic Test-02.pdf + CN Topic Test-01.pdf] · Medium

Q199. Which layer protocols provide user and application programme support such as passwords, resource sharing and file transfer?

किस layer के protocols उपयोगकर्ता और application programme को passwords, resource sharing और file transfer जैसी सहायता देते हैं?

(A) Layer 2 protocols **(B)** Layer 5 protocols
(C) Layer 7 protocols **(D)** Layer 10 protocols

[Source: CN Topic Test-02.pdf] · Medium

Q200. Which of the following protocols is primarily used for secure web browsing?

निम्न में से कौन-सा protocol मुख्य रूप से सुरक्षित web browsing के लिए उपयोग होता है?

(A) FTP **(B)** HTTP
(C) HTTPS **(D)** SMTP

[Source: CN Topic Test-01.pdf] · Medium

Q201. Which protocol is generally used to encrypt data between a web browser and a web server?

Web browser और web server के बीच data encrypt करने के लिए आमतौर पर कौन-सा protocol उपयोग होता है?

(A) SSL/TLS **(B)** Digital Signature
(C) IPSec **(D)** TCP

[Source: CN Topic Test-01.pdf] · Medium

Q202. Which protocol is used for secure shell (remote) login?

Secure shell (remote) login के लिए कौन-सा protocol उपयोग होता है?

(A) FTP **(B)** SSH
(C) Telnet **(D)** SNMP

[Source: CN Topic Test 6.pdf + Medium 30 CN MCQ bank] · Medium

Q203. A firewall blocks all ports except 22. What service remains accessible?

एक firewall port 22 को छोड़कर सब बंद कर देता है. कौन-सी सेवा अभी भी खुली है?

(A) FTP **(B)** SSH
(C) HTTPS **(D)** SMTP

[Source: BCI Subject Test-8.pdf] · Medium

Q204. Which port does Telnet use by default?

Telnet डिफ़ॉल्ट रूप से कौन-सा port उपयोग करता है?

(A) 21 **(B)** 23
(C) 22 **(D)** 25

[Source: CN Topic Test 6.pdf] · Medium

Q205. Which of the following uses port number 25?

निम्न में से कौन-सा port number 25 उपयोग करता है?

(A) FTP **(B)** HTTP
(C) SMTP **(D)** Telnet

[Source: CN Topic Test 6.pdf] · Medium

Q206. Which protocol uses different ports for control and data connections?

कौन-सा protocol control और data connections के लिए अलग-अलग ports उपयोग करता है?

(A) FTP **(B)** HTTP
(C) DNS **(D)** Telnet

[Source: BCI_SCI Subject Test-7.pdf] · Medium

Q207. Which protocol is used for file transfer over a network?

Network पर file transfer के लिए कौन-सा protocol उपयोग होता है?

(A) FTP **(B)** HTTP
(C) DNS **(D)** ARP

[Source: CN Topic Test 6.pdf + Medium 30 CN MCQ bank] · Medium

Q208. Which application allows a user to access and use remote files without actually transferring them?

कौन-सा application उपयोगकर्ता को remote files को बिना वास्तव में स्थानांतरित किए उपयोग करने देता है?

(A) DNS **(B)** FTP
(C) NFS **(D)** Telnet

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

Q209. Which protocol translates domain names to IP addresses?

कौन-सा protocol domain names को IP addresses में बदलता है?

(A) DHCP **(B)** DNS
(C) HTTP **(D)** FTP

[Source: CN Topic Test 5.pdf + CN Topic Test-01.pdf] · Medium

Q210. What does a 'ping' command test?

'ping' command क्या जाँचता है?

(A) IP configuration **(B)** Host reachability
(C) DNS resolution **(D)** Port scanning

[Source: CN Topic Test 6.pdf] · Medium

Q211. Which HTTP status code means 'Forbidden'?

कौन-सा HTTP status code का अर्थ है 'Forbidden'?

(A) 200 **(B)** 301
(C) 403 **(D)** 404

[Source: Oliveboard SSC CGL Tier-2 bank] · Medium

Q212. Lightweight Directory Access Protocol (LDAP) is used for:

Lightweight Directory Access Protocol (LDAP) किस काम आता है?

(A) Routing the packets **(B)** Authentication and directory access
(C) Obtaining an IP address **(D)** Domain name resolving

[Source: Unit Test-4 Solution.pdf (OCR), p29] · Medium

Q213. Which protocol is used for receiving an email?

Email प्राप्त करने के लिए कौन-सा protocol उपयोग होता है?

(A) HTTP **(B)** FTP
(C) POP3 **(D)** SSH

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

Q214. Which of the following is the odd one out?

निम्न में से असंगत कौन-सा है?

(A) IMAP **(B)** POP
(C) SNMP **(D)** SMTP

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

Q215. ICMP is a network-layer protocol used by devices to diagnose network issues. ICMP stands for:

ICMP network-layer protocol है जो devices को network समस्याएँ जाँचने में सहायता करता है. ICMP का पूरा नाम है:

(A) Internet Control Manage Process **(B)** Internet Comfort Mass Protocol
(C) Information Control Manage Protocol **(D)** Internet Control Message Protocol

[Source: IBPS PO Computer Awareness PYQ (via Testbook solved)] · Medium

Q216. POP3 stands for:

POP3 का पूरा नाम है:

(A) Peer to Peer Protocol **(B)** Peer to Post Protocol
(C) Peer Office Protocol **(D)** Post Office Protocol

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Medium

Q217. SLIP stands for:

SLIP का पूरा नाम है:

(A) System Line Internet Protocol **(B)** Serial Line Internet Protocol
(C) Signal Line Internet Protocol **(D)** Signal Internet Protocol

[Source: IBPS/banking computer banks (via tpointtech MCQ)] · Medium

Q218. Which protocol is used to retrieve e-mails and keep them on the server as well?

कौन-सा protocol e-mail लाता है और उन्हें server पर भी रखता है?

(A) SMTP **(B)** POP3
(C) IMAP **(D)** FTP

[Source: Practice — original (textbook-based)] · Medium

Q219. Translation between ASCII and EBCDIC is performed by the:

ASCII और EBCDIC के बीच अनुवाद करती है:

(A) Application layer **(B)** Presentation layer
(C) Data link layer **(D)** Physical layer

[Source: Practice — original (textbook-based)] · Medium

Q220. HTTPS uses which protocol for security?

HTTPS सुरक्षा के लिए कौन-सा protocol प्रयोग करता है?

(A) SSH **(B)** SSL/TLS
(C) SNMP **(D)** ICMP

[Source: Practice — original (textbook-based)] · Medium

Q221. DHCP is used to:

DHCP प्रयोग होता है:

(A) Resolve names to IPs / नामों को IP में बदलने के लिए **(B)** Assign IP configuration automatically / IP configuration स्वतः देने के लिए
(C) Encrypt e-mails / E-mail encrypt करने के लिए **(D)** Route packets / Packets route करने के लिए

[Source: Practice — original (textbook-based)] · Medium

Q222. In FTP, the control connection uses port number:

FTP में control connection किस port पर चलती है?

(A) 20 **(B)** 21
(C) 25 **(D)** 80

[Source: Practice — original (textbook-based)] · Medium

Q223. Which of the following is a stateless protocol?

निम्न में से कौन stateless protocol है?

(A) FTP **(B)** HTTP
(C) Telnet **(D)** TCP

[Source: Practice — original (textbook-based)] · Medium

Q224. DHCP uses which pair of UDP ports (server port, client port)?

DHCP कौन-सी UDP ports की जोड़ी प्रयोग करता है (server port, client port)?

(A) 67 and 68 / 67 और 68 **(B)** 69 and 70 / 69 और 70
(C) 53 and 54 / 53 और 54 **(D)** 20 and 21 / 20 और 21

[Source: Practice — original (textbook-based)] · Medium

Q225. The first line of an HTTP request message is called the:

HTTP request message की पहली पंक्ति कहलाती है:

(A) Request line **(B)** Header line
(C) Status line **(D)** Body line

[Source: Practice — original (textbook-based)] · Medium

Q226. By default, HTTP/1.1 connections are:

डिफ़ॉल्ट रूप से, HTTP/1.1 connections होते हैं:

(A) Persistent (keep-alive) **(B)** Non-persistent
(C) Simplex **(D)** Datagram-based

[Source: Practice — original (textbook-based)] · Medium

Q227. In HTTP, a conditional GET (using If-Modified-Since) allows:

HTTP में conditional GET (If-Modified-Since के साथ) संभव बनाता है:

(A) Cache validation — the server sends a 304 Not Modified instead of the object if unchanged / Cache validation — अपरिवर्तित होने पर server वस्तु के बजाय 304 Not Modified भेजे **(B)** Automatic language translation / स्वतः भाषा अनुवाद
(C) Password-free login / बिना password login **(D)** Parallel downloads / समांतर downloads

[Source: Practice — original (textbook-based)] · Medium

Q228. The functionalities of the presentation layer include:

Presentation layer के कार्यों में शामिल हैं:

(A) Data compression **(B)** Data encryption
(C) Data description (translation) **(D)** All of the mentioned

[Source: CN Topic Test-02.pdf] · Hard

Q229. Which of the following protocols is NOT secure?

निम्न में से कौन-सा protocol सुरक्षित नहीं है?

(A) HTTPS **(B)** FTP
(C) SSH **(D)** SFTP

[Source: CN Topic Test 6.pdf] · Hard

Q230. Which of the following protocols is used for transferring email messages from one machine to another?

निम्न में से कौन-सा protocol email संदेशों को एक machine से दूसरी में भेजने के लिए उपयोग होता है?

(A) SNMP **(B)** FTP
(C) TELNET **(D)** SMTP

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q231. In one of the pairs below, both protocols can use multiple TCP connections between the same client and the same server. Which pair is it?

नीचे दिए जोड़ों में से किसमें दोनों protocols एक ही client और server के बीच कई TCP connections उपयोग कर सकते हैं?

(A) HTTP, FTP **(B)** HTTP, TELNET
(C) FTP, SMTP **(D)** HTTP, SMTP

[Source: CN Topic Test-02.pdf] · Hard

Q232. Identify the true statements: (1) HTTP may use different TCP connections for different objects of a webpage if non-persistent connections are used (2) FTP uses two TCP connections, one for data and another for control (3) TELNET and FTP can only use TWO connections at a time

सही कथन पहचानें: (1) non-persistent connections में HTTP webpage के अलग-अलग objects के लिए अलग TCP connections उपयोग कर सकता है (2) FTP दो TCP connections उपयोग करता है, एक data के लिए और दूसरा control के लिए (3) TELNET और FTP एक समय में केवल TWO connections ही उपयोग कर सकते हैं

(A) (1) only **(B)** (1) and (2)
(C) (2) and (3) **(D)** (1), (2) and (3)

[Source: CN Topic Test-03.pdf] · Hard

Q233. Which of the following is NOT an application layer protocol?

निम्न में से कौन-सा application layer protocol नहीं है?

(A) HTTP **(B)** IP
(C) TELNET **(D)** FTP

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Hard

Q234. Which is the FALSE statement?

कौन-सा कथन गलत है?

(A) Telnet is used for remote login **(B)** FTP can transfer files between two hosts
(C) TFTP relies on TCP **(D)** SMTP is used to send mail messages

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Hard

Q235. Match TCP/IP layers and protocols — (P) Transport layer (Q) Network (internet) layer (R) Application layer; (1) ICMP (2) FTP (3) UDP. Which matching is correct?

TCP/IP layers और protocols मिलाएँ — (P) Transport layer (Q) Network (internet) layer (R) Application layer; (1) ICMP (2) FTP (3) UDP. कौन-सा मिलान सही है?

(A) P-2, Q-1, R-3 **(B)** P-1, Q-2, R-3
(C) P-3, Q-1, R-2 **(D)** P-3, Q-2, R-1

[Source: Random Test- Rajasthan PYQ.pdf (OCR) + answer_keys.py Q23] · Hard

Q236. Which one of the following is NOT a client-server application?

निम्न में से कौन-सा client-server application नहीं है?

(A) Internet Chat **(B)** Web browsing
(C) E-mail **(D)** Ping

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

Q237. Which of the following is/are example(s) of stateful application layer protocols? I. HTTP II. FTP III. TCP IV. POP3

निम्न में से कौन-सा stateful application layer protocol का उदाहरण है? I. HTTP II. FTP III. TCP IV. POP3

(A) I and II only / केवल I और II **(B)** II only / केवल II
(C) I only / केवल I **(D)** II and III only / केवल II और III

[Source: GATE CS 2016 Set-1] · Hard

Q238. Identify the correct sequence in which the following packets are transmitted by a host when a browser requests a web page from a remote server, assuming that the host has just been restarted:

मान लीजिए host अभी-अभी restart हुआ है और browser एक remote server से web page माँगता है। निम्न packets सही क्रम में transmit होंगे:

(A) HTTP GET request, DNS query, TCP SYN **(B)** DNS query, HTTP GET request, TCP SYN
(C) DNS query, TCP SYN, HTTP GET request **(D)** TCP SYN, DNS query, HTTP GET request

[Source: GATE CS 2016 Set-2] · Hard

Q239. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

ई-मेल भेजने और प्राप्त करने (इसी क्रम में) के लिए निम्न में से किस protocol जोड़ी का प्रयोग हो सकता है?

(A) SMTP and POP3 / SMTP और POP3 **(B)** POP3 and SMTP / POP3 और SMTP
(C) IMAP and SMTP / IMAP और SMTP **(D)** SMTP and IMAP only / केवल SMTP और IMAP

[Source: GATE CS 2019] · Hard

Q240. Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10^9 bits/second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

एक लंबी TCP session की bandwidth 1 Gbps (= 10^9 bits/second) है और वह sequence number 1234 से शुरू होती है। इस sequence number के पुनः प्रयोग से पहले न्यूनतम समय (seconds में, निकटतम पूर्णांक) कितना है?

(A) 34 **(B)** 35
(C) 43 **(D)** 32

[Source: GATE CS 2018] · Hard

Q241. In the TCP three-way handshake, host P sends a connection request to Q with SYN = 1, SEQ = X, ACK = 0. Q accepts the connection. Which of the following represents the information in the TCP segment header sent by Q to P?

TCP three-way handshake में host P, Q को connection request भेजता है: SYN = 1, SEQ = X, ACK = 0। Q connection स्वीकार करता है। Q द्वारा P को भेजे segment header में क्या होगा?

(A) SYN = 1, SEQ = Y, ACK = X **(B)** SYN = 1, SEQ = X, ACK = Y
(C) SYN = 1, SEQ = Y, ACK = X+1 **(D)** SYN = 0, SEQ = Y, ACK = X+1

[Source: GATE CS 2021 Set-2] · Hard

Q242. Consider data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL, is _____.

1 Gbps link पर TCP से data transfer हो रहा है। Maximum segment lifetime (MSL) 60 seconds है। MSL के दौरान sequence number space के wrap होने से बचने के लिए TCP header के sequence number field में न्यूनतम कितने bits चाहिए?

(A) 32 **(B)** 33
(C) 34 **(D)** 36

[Source: GATE CS 2022] · Hard

Q243. A TCP client and server complete data transfer. The client calls close; a FIN segment is sent to the server, which responds with an ACK received by the client. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server?

TCP client-server का data transfer पूरा हुआ। Client close करता है; server को FIN जाता है; server का ACK client को मिलता है। RFC 793 की state diagram के अनुसार client-side connection server के FIN की प्रतीक्षा किस state में करती है?

(A) TIME_WAIT **(B)** FIN_WAIT_2
(C) CLOSE_WAIT **(D)** LAST_ACK

[Source: GATE CS 2017 Set-1] · Hard

Q244. In the TCP three-way handshake, let the three packets exchanged during connection establishment be P1, P2 and P3 in order. Which of the following is TRUE about the TCP header flags set in these packets?

TCP three-way handshake में connection establishment के दौरान तीन packets P1, P2 और P3 (इसी क्रम में) बदले जाते हैं। इनमें सेट TCP header flags के बारे में क्या सही है?

(A) P1: SYN; P2: SYN, ACK; P3: ACK **(B)** P1: SYN, ACK; P2: ACK; P3: FIN
(C) P1: SYN; P2: ACK; P3: FIN **(D)** P1: ACK; P2: SYN; P3: ACK

[Source: GATE CS 2025 Set-1] · Hard

Q245. The World Wide Web (WWW) is:

World Wide Web (WWW) क्या है?

(A) another name for the Internet / Internet का दूसरा नाम
(B) a worldwide connection of computers / कंप्यूटरों का विश्वव्यापी जुड़ाव
(C) a worldwide collection of information / सूचना का विश्वव्यापी संग्रह **(D)** a collection of linked information residing on computers connected by the Internet / Internet से जुड़े कंप्यूटरों पर रखी link की गई सूचनाओं का संग्रह

[Source: EMRS PGT Computer Science] · Hard

Q246. In the DHCP DORA process, if two DHCP servers send OFFERs, the client usually accepts:

DHCP DORA process में यदि दो DHCP servers OFFER भेजें, तो client सामान्यतः स्वीकार करता है:

(A) The first OFFER received / सबसे पहले प्राप्त OFFER **(B)** The OFFER with the longest lease / सबसे लंबी lease वाला OFFER
(C) Both and merges them / दोनों, उन्हें मिलाकर **(D)** Neither — it uses a static IP / कोई नहीं — static IP लेता है

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Hard

Q247. Which of the following is NOT a valid HTTP status code?

निम्न में से कौन-सा वैध HTTP status code नहीं है?

(A) 250 **(B)** 200
(C) 404 **(D)** 503

[Source: Practice — original (textbook-based)] · Hard

Answer Key — sirf letters (no solutions)

Q1: B **Q2:** A **Q3:** C **Q4:** B **Q5:** C **Q6:** A **Q7:** C **Q8:** A **Q9:** D **Q10:** D
Q11: A **Q12:** A **Q13:** B **Q14:** C **Q15:** A **Q16:** B **Q17:** B **Q18:** B **Q19:** B **Q20:** C
Q21: B **Q22:** B **Q23:** B **Q24:** B **Q25:** A **Q26:** A **Q27:** D **Q28:** C **Q29:** A **Q30:** B
Q31: C **Q32:** B **Q33:** B **Q34:** B **Q35:** B **Q36:** B **Q37:** B **Q38:** A **Q39:** A **Q40:** A
Q41: D **Q42:** C **Q43:** D **Q44:** B **Q45:** A **Q46:** C **Q47:** D **Q48:** C **Q49:** A **Q50:** B
Q51: C **Q52:** B **Q53:** C **Q54:** C **Q55:** D **Q56:** B **Q57:** D **Q58:** C **Q59:** A **Q60:** B
Q61: B **Q62:** A **Q63:** C **Q64:** B **Q65:** B **Q66:** D **Q67:** C **Q68:** C **Q69:** A **Q70:** C
Q71: B **Q72:** B **Q73:** C **Q74:** D **Q75:** B **Q76:** B **Q77:** C **Q78:** C **Q79:** B **Q80:** B
Q81: B **Q82:** C **Q83:** A **Q84:** B **Q85:** A **Q86:** D **Q87:** A **Q88:** A **Q89:** B **Q90:** B
Q91: A **Q92:** B **Q93:** A **Q94:** A **Q95:** B **Q96:** C **Q97:** C **Q98:** C **Q99:** C **Q100:** B
Q101: B **Q102:** A **Q103:** B **Q104:** B **Q105:** A **Q106:** C **Q107:** B **Q108:** B **Q109:** B **Q110:** B
Q111: C **Q112:** B **Q113:** C **Q114:** B **Q115:** B **Q116:** A **Q117:** A **Q118:** A **Q119:** A **Q120:** A
Q121: A **Q122:** D **Q123:** C **Q124:** C **Q125:** C **Q126:** A **Q127:** B **Q128:** A **Q129:** C **Q130:** C
Q131: C **Q132:** D **Q133:** D **Q134:** B **Q135:** A **Q136:** D **Q137:** B **Q138:** C **Q139:** C **Q140:** A
Q141: A **Q142:** A **Q143:** A **Q144:** A **Q145:** A **Q146:** A **Q147:** A **Q148:** A **Q149:** B **Q150:** C
Q151: B **Q152:** C **Q153:** B **Q154:** A **Q155:** C **Q156:** A **Q157:** C **Q158:** B **Q159:** D **Q160:** C
Q161: C **Q162:** D **Q163:** B **Q164:** A **Q165:** D **Q166:** B **Q167:** B **Q168:** C **Q169:** A **Q170:** C
Q171: D **Q172:** C **Q173:** A **Q174:** C **Q175:** C **Q176:** D **Q177:** A **Q178:** B **Q179:** D **Q180:** B
Q181: C **Q182:** C **Q183:** C **Q184:** A **Q185:** D **Q186:** A **Q187:** B **Q188:** D **Q189:** B **Q190:** A
Q191: B **Q192:** A **Q193:** A **Q194:** A **Q195:** B **Q196:** C **Q197:** A **Q198:** A **Q199:** C **Q200:** C
Q201: A **Q202:** B **Q203:** B **Q204:** B **Q205:** C **Q206:** A **Q207:** A **Q208:** C **Q209:** B **Q210:** B
Q211: C **Q212:** B **Q213:** C **Q214:** C **Q215:** D **Q216:** D **Q217:** B **Q218:** C **Q219:** B **Q220:** B
Q221: B **Q222:** B **Q223:** B **Q224:** A **Q225:** A **Q226:** A **Q227:** A **Q228:** D **Q229:** B **Q230:** D
Q231: A **Q232:** B **Q233:** B **Q234:** C **Q235:** C **Q236:** D **Q237:** B **Q238:** C **Q239:** A **Q240:** A
Q241: C **Q242:** B **Q243:** B **Q244:** A **Q245:** D **Q246:** A **Q247:** A